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### **METHOD FOR OPTICAL EMISSION SPECTROSCOPY (OES) DETECTOR SIGNAL SENSITIVITY IMPROVEMENT**

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#### **Abstract**

Aspects of the present disclosure provide a method for optical emission spectroscopy (OES) detector signal sensitivity improvement. For example, the method can include scanning OES spectra of an optical emission, and defining all wavelengths of the optical emission whose intensities do not satisfy a predetermined requirement. The method can further include adjusting the intensity of each of the defined wavelengths by a first predetermined factor, and calculating an amplification/attenuation coefficient corresponding to the intensity of the defined wavelength based on the first predetermined factor. The method can further include conducting data analysis on the optical emission by taking into account all the amplification/attenuation coefficients.

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## Background/Summary

### FIELD OF THE INVENTION

[0001] The present disclosure relates to improvement to optical emission spectroscopy (OES) detector signal sensitivity.

### BACKGROUND

[0002] The background description provided herein is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent the work is described in this background section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

[0003] Optical emission spectroscopy (OES) enables analysis of elemental composition of samples. The intensities of spectra lines of OES spectra across one or more wavelengths may be too large to be within a detection range or too small to be detected.

### SUMMARY

[0004] Aspects of the present disclosure provide a method for optical emission spectroscopy (OES) detector signal sensitivity improvement. For example, the method can include scanning OES spectra of an optical emission and defining all wavelengths of the optical emission whose intensities do not satisfy a predetermined requirement. The method can further include adjusting the intensity of each of the defined wavelengths by a first predetermined factor and calculating an amplification/attenuation coefficient corresponding to the intensity of the defined wavelength based on the first predetermined factor. The method can further include conducting data analysis on the optical emission by taking into account all the amplification/attenuation coefficients.

[0005] In an embodiment, the method can further include adjusting the intensity of each of the defined wavelengths, which is adjusted by the first predetermined factor, by a second predetermined factor if the intensity of the defined wavelength still does not satisfy the predetermined requirement, wherein the amplification/attenuation coefficient is calculated based on the first predetermined factor and the second predetermined factor. In some embodiments, the amplification/attenuation coefficient can be calculated by multiplying the first predetermined factor and the second predetermined factor. In an embodiment, the first predetermined factor can be equal to the second predetermined factor. In another embodiment, the first predetermined factor can be different from the second predetermined factor. For example, the first predetermined factor can be greater than the second predetermined factor.

[0006] In an embodiment, the predetermined requirement can include a maximum threshold, and adjusting the intensity of each of the defined wavelengths by the predetermined factor can include reducing the intensity of each of the defined wavelengths by the predetermined factor. For example, the maximum threshold can correspond to a greatest one of intensities within a detection range of the OES spectra. As another example, the maximum threshold can correspond to 80% of a greatest one of intensities within a detection range of the OES spectra. In another embodiment, the predetermined requirement can include a minimum threshold, and adjusting the intensity of each of the defined wavelengths by the predetermined factor can include increasing the intensity of each of the defined wavelengths by the predetermined factor.

[0007] Aspects of the present disclosure also provide an apparatus for OES detector signal sensitivity improvement. For example, the apparatus can include an OES detector configured to scan OES spectra of an optical emission and define all wavelengths of the optical emission whose intensities do not satisfy a predetermined requirement. The apparatus can further include a gain adjusting module coupled to the OES detector, the gain adjusting module configured to adjust the intensity of each of the defined wavelengths by a first predetermined factor. The apparatus can

further include an analog-to-digital converter (ADC) coupled to the gain adjusting module, the ADC configured to convert the intensity of each of the defined wavelengths from an analog format to a digital format. The apparatus can further include a controller coupled to the ADC and the gain adjusting module, the controller configured to determine whether the intensity of each of the defined wavelengths in the digital format satisfies a predetermined requirement, generate the first predetermined factor if the intensity of the defined wavelength does not satisfy the predetermined requirement, calculate an amplification/attenuation coefficient corresponding to the intensity of the defined wavelength based on the first predetermined factor, and conduct data analysis on the optical emission by taking into account all the amplification/attenuation coefficients.

[0008] In an embodiment, the controller can be further configured to adjust the intensity of each of the defined wavelengths, which is adjusted by the first predetermined factor, by a second predetermined factor if the intensity of the defined wavelength still does not satisfy the predetermined requirement, and calculate the amplification/attenuation coefficient based on the first predetermined factor and the second predetermined factor. In some embodiments, the controller can be configured to calculate the amplification/attenuation coefficient by multiplying the first predetermined factor and the second predetermined factor.

[0009] In an embodiment, the predetermined requirement can include a maximum threshold, and the gain adjusting module can be configured to adjust the intensity of each of the defined wavelengths by reducing the intensity of each of the defined wavelengths by the predetermined factor. In another embodiment, the predetermined requirement can include a maximum threshold, and the gain adjusting module can be configured to adjust the intensity of each of the defined wavelengths by increasing the intensity of each of the defined wavelengths by the predetermined factor.

[0010] Note that this summary section does not specify every embodiment and/or incrementally novel aspect of the present disclosure or claimed invention. Instead, this summary only provides a preliminary discussion of different embodiments and corresponding points of novelty. For additional details and/or possible perspectives of the invention and embodiments, the reader is directed to the Detailed Description section and corresponding figures of the present disclosure as further discussed below.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0011] Various embodiments of this disclosure that are proposed as examples will be described in detail with reference to the following figures, wherein like numerals reference like elements, and wherein:

[0012] FIG. 1 is a schematic diagram of an optical analyzer apparatus that makes use of optical emission spectroscopy (OES) according to some embodiments of the present disclosure;

[0013] FIG. 2 is a schematic diagram illustrating an exemplary spectrometer system according to some embodiments of the present disclosure;

[0014] FIG. 3 is a spectra diagram illustrating the intensity distribution of light that is analyzed by a spectrometer system across a range of wavelengths;

[0015] FIG. 4 is a functional block diagram of an exemplary optical analyzer apparatus for OES detector signal sensitivity improvement according to some embodiments of the present disclosure;

[0016] FIG. 5 is a flow chart illustrating an exemplary method for OES detector signal sensitivity improvement according to some embodiments of the present disclosure; and

[0017] FIG. 6 is a spectra diagram illustrating the intensity distribution of light that is analyzed by the optical analyzer apparatus of FIG. 4 across a range of wavelengths according to some embodiments of the present disclosure.

#### DETAILED DESCRIPTION

[0018] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed. Further, spatially relative terms, such as “top,” “bottom,” “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0019] The order of discussion of the different steps as described herein has been presented for clarity sake. In general, these steps can be performed in any suitable order. Additionally, although each of the different features, techniques, configurations, etc. herein may be discussed in different places of this disclosure, it is intended that each of the concepts can be executed independently of each other or in combination with each other. Accordingly, the present invention can be embodied and viewed in many different ways.

[0020] Optical emission spectroscopy (OES) is a reliable and extensively used elemental analysis technique used to analyze the elemental composition of samples, such as metals and alloys, and to detect the presence and relative concentrations of various gas species in a process chamber, in which deposition and etching, for example, are performed in a plasma environment. OES involves the collection, spectral dispersion, and detection of light. FIG. 1 is a schematic diagram of an optical analyzer apparatus **10** (or an OES system) that makes use of OES according to some embodiments of the present disclosure. The optical analyzer apparatus **10** can include an excitation source **11**, an optical assembly optically coupled to the excitation source **11**, a spectrometer system **13** optically coupled to the optical assembly **12**, and a controller **15** electrically coupled to the excitation source **11** and the spectrometer system **13**. In an embodiment, the controller **15** can be provided by a computer that includes a processor and a memory, the memory being configured to store computer instructions, programs and/or algorithms that, when executed by the processor, causes to the computer to operate as the controller **105** according to some embodiments of the present disclosure.

[0021] The excitation source **11** can be controlled by the controller **15** to generate tunable excitation **11a** to invoke an optical emission **14a** emitted from a sample **14**. In an embodiment, the excitation source **11** can include a laser source that is configured to generate a single laser pulse of a series of laser pulses of high peak power that can be focused by the optical assembly **12** to the sample **14** to form a plasma plume on the surface of the sample **14**. During the plasma formation, free electrons start to recombine with ions, which invokes from the sample **14** the optical emission at wavelength(s) that characterize the elements on the surface of the sample **14**.

[0022] The optical assembly **12** can be configured to focus the excitation **11a** onto the surface of the sample **14** and transferring the optical emission **14a** emitted from the sample **14** into the spectrometer system **13**. In an embodiment, the optical assembly **12** can include one or more collecting and focusing lens and mirrors **12a** that are configured to collect and focus the excitation **11a** generated by the excitation source **11**, e.g., the laser pulses generated by the laser source, onto

the surface of the sample **14**. In another embodiment, the optical assembly **12** can further include one or more focusing lens and mirrors **12b** that are configured to focus the optical emission **14a** into the spectrometer system **13**, e.g., via an input fiber cable (not shown) and a connector **13a** to which the fiber cable is secured. In some embodiments, the optical assembly **12** can further include a filter **13c**, e.g., a notch filter that is configured to attenuate frequencies within a specific range while passing all other frequencies unaltered.

[0023] The spectrometer system **13** can be configured to determine the elemental composition of the sample **14** according to the optical emission **14a** transferred thereinto, for example, by analyzing the intensity distribution of the dispersed optical emission **14a** across a range of wavelengths.

[0024] FIG. **2** is a schematic diagram illustrating an exemplary spectrometer system **20**, e.g., the spectrometer system **13**, according to some embodiments of the present disclosure. A connector **110** can secure an input fiber cable (not shown) to a spectrometer **100**. For example, the connector **110** can be a subminiature version A (SMA) connector. As another example, the input fiber can be an UV/DUV single core fiber cable. The input fiber can couple the light **190** generated from a light source, e.g., the optical emission **14a** emitted from the sample **14** shown in FIG. **1**, to the spectrometer **100**.

[0025] The spectrometer **100** can be used to determine spectral characteristics about an object and/or element through analysis of its interactions with light or of the spectral components (or spectral lines) of the light itself. The spectrometer can be configured to separate and measure the distinct spectral components of light. There are two kinds of spectrometers according to different light-collecting techniques. One is a scanning spectrometer, which can collect light exiting from an exit slit by using a single-channel detector, such as photomultiplier tube (PMT) detector and silicon detector, that is positioned behind the exit slit. The scanning spectrometer may include a diffraction grating, which is driven by a stepper motor to rotate in steps of predetermined degrees (scanning step number) such that light of a specific wavelength can reach the exit slit and be detected by the PMT detector. In another embodiment, a scanning spectrometer can include a hyperspectral imaging system, where wavelength scanning is accomplished by scanning the passband of an optical filter placed in front on an imaging device. For example, the hyperspectral imaging system can include collecting optics, a wavelength tunable filter (i.e., the optical filter), and an array detector (i.e., the imaging system). The collecting optics can collect an optical signal, the wavelength tunable filter can tune the wavelengths of the collected optical signal such that the tuned optical signal can have certain scanning step passband wavelength numbers, and the tuned optical signal can be projected onto the array detector. An exemplary hyperspectral imaging system is described in US 2020/0372629 A1 and US 2020/0373210 A1, which are incorporated therein for reference in their entirety. The other kind of spectrometers is a charge coupled device (CCD) spectrometer, which can collect light from a fixed diffraction grating by using a multi-channel CCD, or other type of detector, e.g. a CMOS detector, photodiode array, etc. Accordingly, the CCD spectrometer can collect the whole spectrum of the light simultaneously.

[0026] A slit **120** can be mounted behind the connector **110**. The slit **120** can control the amount of the light **190** (i.e., photon flux) that enters the spectrometer **100** and limit the spatial distribution of the light **190**. The slit **120** typically is rectangular and on the order of the wavelength of the light to be measured, e.g., the light **190**. For example, the slit **120** can be sized from 5  $\mu\text{m}$  to 800  $\mu\text{m}$  in width and 1 mm to 2 mm in height. After passing through the slit **120**, the light **190** will be redefined with a Voigt function that is the convolution of the Lorentzian function and an instrumental broadening function, which is attributable to the non-negligible width of the slit **120** and is typically a Gaussian function. The width of the slit **120** can thus affect the spectral resolution (or the spectral bandpass (BP)) of the spectrometer **100**; the narrower the slit **120**, the higher the spectral resolution. However, the slit **120** that is narrow can decrease the strength and signal-to-noise ratio (SNR) of the light **190** that enters the spectrometer **100**. In an embodiment, the SNR of

the light **190** can be increased by averaging multiple measurements or by a moving average method.

[0027] A collimating optic **130** can ensure that the light **190** from the slit **120** travels toward a diffraction grating **140** in a parallel, collimated fashion because the diffraction grating **140** can only work correctly if it is illuminated with collimated light. For example, the collimating optic **130** can be a lens, which can be disposed along an optical path from the slit **120** to the diffraction grating **140**. As another example, the collimating optic **130** can be a mirror (e.g., an off-line spherical mirror), which can be disposed beside the slit **120** and the diffraction grating **140**, and be disposed such that the slit **120** is located at its focal point.

[0028] The diffraction grating **140** can diffract (or separate) the collimated light **190** from the collimating optic **130** into a plurality of spectral components, e.g., the first to third spectral components **191** to **193**, and direct the separated first to third spectral components **191** to **193** onto a focusing optic **150** at different positions. The first to third spectral components **191** to **193**, which have different wavelengths, can leave the diffraction grating **140** at different angles, but a bundle of light of each of the first to third spectral components **191** to **193** can leave the diffraction grating **140** at the same angle. The groove density (or groove frequency) of the diffraction grating **140**, i.e., the amount of grooves per mm, can determine the ability (i.e., the spectral BP and spectral resolution) of the spectrometer **100** to discriminate between adjacent spectral components, e.g., the first to third spectral components **191** to **193**. The groove facet angle (or blaze angle) of the diffraction grating **140** can determine the overall shapes of the first to third spectral components **191** to **193**. In an embodiment, the diffraction grating **140** can be blazed to provide high diffraction efficiency at a specific wavelength, i.e., a blaze wavelength ( $\lambda_{\text{sub.B}}$ ). The diffraction efficiency will decrease by 50% at  $0.6 \times \lambda_{\text{sub.B}}$  and  $1.8 \times \lambda_{\text{sub.B}}$ . The SNR of the spectrometer **100** can be improved by biasing the blaze wavelength of the diffraction grating **140** toward the weak side of the spectral range.

[0029] In an embodiment, the diffraction grating **140** can be transmissive, like a multi-slit aperture, and be disposed along an optical path from the collimating optic **130** to the focusing optic **150**. In another embodiment, the diffraction grating **140** can be reflective, with its grooved surface (representing the multi-slit aperture) coated with a highly reflective material, such as aluminum, and be disposed beside the collimating optic **130** and the focusing optic **150**.

[0030] In an embodiment, the diffraction grating **140** can be a ruled grating, which is formed by etching a plurality of parallel, angled grooves onto the surface of a substrate and coating the grooves with a highly reflective material. In another embodiment, the diffraction grating **140** can be a holographic grating, which is formed by interfering two UV beams in a fringe field having a standing-wave pattern exposed to a polished substrate coated with photoresist, and thus has sinusoidal cross-sectional grooves. As compared with the ruled grating, the holographic grating can exhibit much less stray light, but is more difficult to be blazed due to its sinusoidal groove profile.

[0031] The focusing optic **150** can focus the separated first to third spectral components **191** to **193** onto a detector **160** at different positions based on their different wavelengths. For example, the focusing optic **150** can focus the first to third spectral components **191** to **193** onto first to third pixels **161** to **163** of the detector **160**, respectively, and the pixel numbers (or pixel positions in nm) of the first to third pixels **161** to **163** can be associated with the wavelengths of the first to third spectral components **191** to **193**. In an embodiment, the focusing optic **150** can be a lens, which can be disposed along an optical path from the diffraction grating **140** to the detector **160**. In another embodiment, the focusing optic **150** can be a mirror, which can be disposed beside the diffraction grating **140** and the detector **160**.

[0032] The detector **160** can be, for example, a CCD, a CMOS sensor, a photodiode array detector, or any other type of array detector. The detector **160** can be arranged on a focal plane of the focusing optic **150**, and collect the first to third spectral components **191** to **193** from the focusing optic **150**. For example, the detector **160** can include a plurality of array-based or linearly arranged

pixels, e.g., the first to third pixels **161** to **163**, which can respond to and interact with the first to third spectral components **191** to **193** that strike them, respectively, and build up their respective charges; the brighter the spectral component (or the higher the intensity of the spectral component) and/or the longer the interaction, the more charge is built. In an embodiment, the detector **160** can include 1,024 linearly arranged pixels. In another embodiment, the detector **160** can include 2,048 linearly arranged pixels. The detector **160** can then output analog signals according to the built charges of the pixels. The analog signals output by the detector **160** may be too high to be directly coupled to an analog-to-digital converter (ADC) **170**. In such a scenario, the analog signals have to be linearly inverted, amplified and shifted before passing to the ADC **170**. The ADC **170** can convert the processed analog signals into digital signals and transmit the digital signals to a controller **172** for further processing. In an embodiment, the controller **172** can include a processor **173** and a memory **174** coupled to the processor **173**. For example, the memory **174**, e.g., a non-transitory computer-readable storage medium, can have stored therein a variety of instructions, programs and algorithms that, when executed by the processor **173**, can cause the controller **172** to perform some associated operations. In an embodiment, the processor **173** and the non-transitory computer-readable storage medium **174** can be installed in an external computer, and the computer can be coupled to the spectrometer **100** via an interface **171**. For example, the interface **171** can be a universal serial bus (USB).

[0033] Optionally, the spectrometer **100** can be equipped with a cooling system **180** if it is to be in measurement of the light **190** with long integration time. For example, the cooling system **180** can be a thermoelectric cooler (TEC). Even when no protons are impinging on the detector **160** and thus no electrons will be excited and no current will be generated accordingly, there are still some electrons that will be excited spontaneously due to their thermal energy, thus generating a dark current. The dark current can accumulate over the integration time, and can affect the measurement of the detector **160** significantly if the integration time is over, for example, 100 ms. Because the dark current comes from thermal fluctuations, lowering the operating temperature of the detector **160** can be a way to decrease the dark current. In an embodiment, the cooling system **180** can be arranged close to the detector **160** to reduce the dark current, thereby reducing the corresponding dark noises and enhancing the dynamic range of the spectrometer **100**.

[0034] The optical analyzer apparatus **10** can scan a broad wavelength spectra, some of spectra lines of which may have low intensities, while the other spectra lines of which may be out of the detection range. FIG. 3 is a spectra diagram **30** illustrating the intensity distribution of light, e.g., the dispersed optical emission **14a**, that is analyzed by a spectrometer system, e.g., the spectrometer system **20**, across a range of wavelengths. As shown, spectra lines of an original wavelength spectra **31** of the optical emission **14a** across first and second wavelength ranges **31a** and **31b** are out of the detection range and their intensities are thus truncated. In order to determine the correct intensities of the spectra lines across the first and second wavelength ranges **31a** and **31b**, the power of laser pulses, e.g., the excitation **11a**, generated by an excitation source, e.g., the excitation source **11**, to invoke the optical emission **14a** to be emitted from the sample **14**, has to be reduced (e.g., becoming 20% of the original intensity), or alternatively or simultaneously the analog signals of the pixels of the detector **160** of the CCD spectrometer **100** have to be attenuated, and the original wavelength spectra **31** can thus be modified to become a modified wavelength spectra **32**, in which the intensities of the spectra lines across the first and second wavelength ranges **31a** and **31b** are within the detection range shown in the spectra diagram **30**.

[0035] As the OES system **10** treats the optical emission **14a** across all the wavelengths equally at hardware (HW) level and the gain/attenuation for all wavelengths is the same, in the modified wavelength spectra **32**, not only the intensities of the spectra lines across the first and second wavelength ranges **31a** and **31b** are reduced, the intensities of the other spectra lines are also reduced, some of which may thus have too small the intensities to be detected. For example, the intensities of the spectra lines of the modified wavelength spectra **32** across third and fourth

wavelength ranges **32a** and **32b**, which are reduced to become 20% of the intensities of the spectra lines of the original wavelength spectra **31**, may be too small to be detected.

[0036] In order to maintain the best signal-to-noise ratio (SNR) and keep the maximum level of sensitivity to wave numbers of interest, aspects of the present disclosure provide mechanisms that selectively amplify (e.g., gain adjustment) or suppress (e.g., attenuation adjustment) of individual spectra lines at hardware (HW) or software (SF) level.

[0037] FIG. **4** is a functional block diagram of an exemplary optical analyzer apparatus **40** (or an OES system) for OES detector signal sensitivity improvement according to some embodiments of the present disclosure. FIG. **5** is a flow chart illustrating an exemplary method **50** for OES detector signal sensitivity improvement according to some embodiments of the present disclosure. In various embodiments, some of the steps of the method **50** shown can be performed concurrently or in a different order than shown, can be substituted by other method steps, or can be omitted. Additional method steps can also be performed as desired. Aspects of the method **50** can be implemented by an optical analyzer apparatus, such as the optical analyzer apparatus **40**.

[0038] The optical analyzer apparatus **40** can be configured to generate an excitation whose power is tunable and/or adjust the gain of analog signals of pixels of a detector of a spectrometer, e.g., the pixels of the detector **160** of the spectrometer **100**, according to whether the intensities of spectra lines emitted from a sample (e.g., various gas species in a process chamber **41**) satisfy a predetermined requirement, such as whether the intensities of the spectra lines exceed an upper threshold or are smaller than a minimum threshold. In an embodiment, various processes, such as deposition and etching, can be performed in the process chamber **41**.

[0039] In an embodiment, the system **400** can include an OES detector **42**, a gain adjusting module **43**, an ADC **44** and a controller **45**. The method **50** can start at step **S51**, at which the OES spectra (including all the spectra lines of the optical emission **14a**) can be scanned, and all wavelengths whose intensities do not satisfy the predetermined requirement can be defined. For example, the spectra lines of the optical emission **14a** whose intensities exceed a maximum threshold, e.g., a greatest one or 80% of the greatest one of intensities within the detection range, can be defined. As another example, the spectra lines of the optical emission **14a** whose intensities are less than a minimum threshold, e.g., 5% of the detection range, can be defined.

[0040] In an embodiment, the OES detector **42** can be configured to scan OES spectra of an optical emission and define all wavelengths of the optical emission whose intensities do not satisfy a predetermined requirement. In an embodiment, the OES detector **42** can include the OES system **10**, and the controller **45** can control the gain adjusting module **43** to adjust the power of the excitation **11a** generated by the excitation source **11** according to the intensities of the spectra lines of the optical emission **14a** across a range of wavelengths. The method **50** can proceed to step **S52**, at which the intensities of the defined wavelengths, which do not satisfy the predetermined requirement, can be adjusted by a predetermined factor, and amplification/attenuation coefficients corresponding to the intensities of the defined wavelengths can be calculated. For example, when the intensity of a first spectra line of the optical emission **14a** across a first wavelength exceeds the maximum threshold, the controller **45** can control the gain adjusting module **43** to reduce the power of the excitation **11a** by a first predetermined factor, e.g., 2. As another example, when the intensity of a second spectra line of the optical emission **14a** across a second wavelength is less than the minimum threshold, the controller **45** can control the gain adjusting module **43** to increase the power of the excitation **11a** by a second predetermined factor, e.g., 2. In another embodiment, the OES detector **42** can include the CCD spectrometer **100**, and the controller **45** can control the gain adjusting module **43** to adjust the gain of the analog signals of the pixels of the detector **160** of the CCD spectrometer **100**. For example, when the intensity of a third spectra line of the optical emission **14a** across a third wavelength exceeds the maximum threshold, the controller **45** can control the gain adjusting module **43** to reduce the gain of (i.e., to attenuate) the analog signals of the pixel by a third predetermined factor, e.g., 2. As another example, when the intensity of a fourth



spectra line of the optical emission **14a** across a fourth wavelength is less than the minimum threshold, the controller **45** can control the gain adjusting module **43** to increase the gain of (i.e., to amplify) the analog signals of the pixels by a fourth predetermined factor, e.g., 2. In various embodiment, the predetermined factor can be included in a pixel-dependent vector such that each wavelength (or pixel) can have its own predetermined factor. The method **50** can proceed to step **S53**.

[0041] At step **S53**, it is determined whether the adjusted intensities of the spectra lines, e.g., the first to fourth spectra lines, of the optical emission **14a** across the defined wavelengths, e.g., the first to fourth wavelengths, satisfy the predetermined requirement. The method **50** can proceed to step **S52** if the adjusted intensities of the spectra lines of the optical emission **14a** across the defined wavelengths still do not satisfy the predetermined requirement, the amplification/attenuation coefficients being calculated by multiplying the multiple predetermined factors; otherwise, the method **50** can proceed to step **S54**, at which the intensities of all the spectra lines of the optical emission **14a** satisfy the predetermined requirement and data analysis can be conducted by taking into account all amplification/attenuation coefficients. In an embodiment, the predetermined factor used for adjusting the intensities of the spectra lines of the optical emission **14a** across the defined wavelengths at step **S52** that is proceeded by the method **50** from step **S51** can be different from or equal to the predetermined factor used for adjusting the intensities of the spectra lines at step **S52** that is proceeded from step **S53**. That is to say, the predetermined factors used for adjusting the intensities of the spectra lines at every iteration of steps **S52** and **S53** can be different from or equal to each other. In an embodiment, the predetermined factor used for adjusting the intensities of the spectra lines at step **S52** executed previously can be greater than the predetermined factor used for adjusting the intensities of the spectra lines at step **S52** executed subsequently.

[0042] FIG. **6** is a spectra diagram **60** illustrating the intensity distribution of light, e.g., the dispersed optical emission **14a**, that is analyzed by a spectrometer system, e.g., the OES system **40**, across a range of wavelengths that includes the first to fourth wavelength ranges **31a**, **31b**, **32a** and **32b** according to some embodiments of the present disclosure. Different from the spectra diagram **30**, which illustrates the original wavelength spectra **31**, the spectra lines of which within the first and second wavelengths **31a** and **31b** have intensities out of the detection range, and the modified wavelength spectra **32**, the spectra lines of which within the third and fourth wavelengths **32a** and **32b** have intensities that are too small to be detected, the spectra diagram **60** illustrates a selectively modified wavelength spectra **61** that includes the spectra lines of the modified wavelength spectra **32** within the first and second wavelength ranges **31a** and **31b**, the intensities of which are out of the detection range initially and are modified to satisfy the predetermined requirement, i.e., less than the maximum threshold, and the spectra lines of the original wavelength spectra **31** within the third and fourth wavelength ranges **32a** and **32b**, the intensities of which are within the detection range and no modification is required to be conducted thereon.

[0043] In the preceding description, specific details have been set forth, such as a particular geometry of a processing system and descriptions of various components and processes used therein. It should be understood, however, that techniques herein may be practiced in other embodiments that depart from these specific details, and that such details are for purposes of explanation and not limitation. Embodiments disclosed herein have been described with reference to the accompanying drawings. Similarly, for purposes of explanation, specific numbers, materials, and configurations have been set forth in order to provide a thorough understanding. Nevertheless, embodiments may be practiced without such specific details. Components having substantially the same functional constructions are denoted by like reference characters, and thus any redundant descriptions may be omitted.

[0044] Various techniques have been described as multiple discrete operations to assist in understanding the various embodiments. The order of description should not be construed as to

imply that these operations are necessarily order dependent. Indeed, these operations need not be performed in the order of presentation. Operations described may be performed in a different order than the described embodiment. Various additional operations may be performed and/or described operations may be omitted in additional embodiments.

[0045] “Substrate” or “target substrate” as used herein generically refers to an object being processed in accordance with the invention. The substrate may include any material portion or structure of a device, particularly a semiconductor or other electronics device, and may, for example, be a base substrate structure, such as a semiconductor wafer, reticle, or a dielectric layer on or overlying a base substrate structure such as a thin film. Thus, substrate is not limited to any particular base structure, underlying dielectric layer or overlying dielectric layer, patterned or unpatterned, but rather, is contemplated to include any such dielectric layer or base structure, and any combination of dielectric layers and/or base structures. The description may reference particular types of substrates, but this is for illustrative purposes only.

[0046] Those skilled in the art will also understand that there can be many variations made to the operations of the techniques explained above while still achieving the same objectives of the invention. Such variations are intended to be covered by the scope of this disclosure. As such, the foregoing descriptions of embodiments of the invention are not intended to be limiting. Rather, any limitations to embodiments of the invention are presented in the following claims.

## Claims

1. A method, comprising: scanning optical emission spectroscopy (OES) spectra of an optical emission, and defining all wavelengths of the optical emission whose intensities do not satisfy a predetermined requirement; adjusting the intensity of each of the defined wavelengths by a first predetermined factor, and calculating an amplification/attenuation coefficient corresponding to the intensity of the defined wavelength based on the first predetermined factor; and conducting data analysis on the optical emission by taking into account all the amplification/attenuation coefficients.
2. The method of claim 1, further comprising: adjusting the intensity of each of the defined wavelengths, which is adjusted by the first predetermined factor, by a second predetermined factor if the intensity of the defined wavelength still does not satisfy the predetermined requirement, wherein the amplification/attenuation coefficient is calculated based on the first predetermined factor and the second predetermined factor.
3. The method of claim 2, wherein the amplification/attenuation coefficient is calculated by multiplying the first predetermined factor and the second predetermined factor.
4. The method of claim 2, wherein the first predetermined factor is equal to the second predetermined factor.
5. The method of claim 2, wherein the first predetermined factor is different from the second predetermined factor.
6. The method of claim 5, wherein the first predetermined factor is greater than the second predetermined factor.
7. The method of claim 1, wherein the predetermined requirement includes a maximum threshold, and adjusting the intensity of each of the defined wavelengths by the predetermined factor includes reducing the intensity of each of the defined wavelengths by the predetermined factor.
8. The method of claim 7, wherein the maximum threshold corresponds to a greatest one of intensities within a detection range of the OES spectra.
9. The method of claim 7, wherein the maximum threshold corresponds to 80% of a greatest one of intensities within a detection range of the OES spectra.
10. The method of claim 1, wherein the predetermined requirement includes a minimum threshold, and adjusting the intensity of each of the defined wavelengths by the predetermined factor includes

increasing the intensity of each of the defined wavelengths by the predetermined factor.

**11.** An apparatus, comprising: an optical emission spectroscopy (OES) detector configured to scan OES spectra of an optical emission and define all wavelengths of the optical emission whose intensities do not satisfy a predetermined requirement; a gain adjusting module coupled to the OES detector, the gain adjusting module configured to adjust the intensity of each of the defined wavelengths by a first predetermined factor; an analog-to-digital converter (ADC) coupled to the gain adjusting module, the ADC configured to convert the intensity of each of the defined wavelengths from an analog format to a digital format; and a controller coupled to the ADC and the gain adjusting module, the controller configured to determine whether the intensity of each of the defined wavelengths in the digital format satisfies a predetermined requirement, generate the first predetermined factor if the intensity of the defined wavelength does not satisfy the predetermined requirement, calculate an amplification/attenuation coefficient corresponding to the intensity of the defined wavelength based on the first predetermined factor, and conduct data analysis on the optical emission by taking into account all the amplification/attenuation coefficients.

**12.** The apparatus of claim 11, wherein the controller is further configured to adjust the intensity of each of the defined wavelengths, which is adjusted by the first predetermined factor, by a second predetermined factor if the intensity of the defined wavelength still does not satisfy the predetermined requirement, and calculate the amplification/attenuation coefficient based on the first predetermined factor and the second predetermined factor.

**13.** The apparatus of claim 12, wherein the controller is configured to calculate the amplification/attenuation coefficient by multiplying the first predetermined factor and the second predetermined factor.

**14.** The apparatus of claim 12, wherein the first predetermined factor is equal to the second predetermined factor.

**15.** The apparatus of claim 12, wherein the first predetermined factor is different from the second predetermined factor.

**16.** The apparatus of claim 15, wherein the first predetermined factor is greater than the second predetermined factor.

**17.** The apparatus of claim 11, wherein the predetermined requirement includes a maximum threshold, and the gain adjusting module is configured to adjust the intensity of each of the defined wavelengths by reducing the intensity of each of the defined wavelengths by the predetermined factor.

**18.** The apparatus of claim 17, wherein the maximum threshold corresponds to a greatest one of intensities within a detection range of the OES spectra.

**19.** The apparatus of claim 17, wherein the maximum threshold corresponds to 80% of a greatest one of intensities within a detection range of the OES spectra.

**20.** The apparatus of claim 11, wherein the predetermined requirement includes a minimum threshold, and the gain adjusting module is configured to adjust the intensity of each of the defined wavelengths by increasing the intensity of each of the defined wavelengths by the predetermined factor.

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